ProvenanceLab

PRODUCT REQUIREMENTS DOCUMENT

Reproducible molecular analysis
with explicit uncertainty and evidence.

HM-01 | Draft for team alignment | Version 1.0

A shareable product specification for the HM-01 challenge

How might we help biotechnology students analyse and compare molecules reproducibly, while clearly identifying invalid inputs and missing evidence?

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CORE PROMISE

Every result must be traceable to a submitted input, an explicit rule, a pinned software environment, and a declared evidence snapshot. If a value cannot be produced, ProvenanceLab reports the reason instead of filling the gap.

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This PRD separates the target product from the current prototype so the team can plan, build, and demonstrate without overstating capability.

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READING GUIDE Sections 1-4 establish product intent. Sections 5-10 define build requirements. Sections 11-15 govern delivery, verification, and team decisions.

1. Executive summary

Biotechnology students frequently obtain molecular structures from papers, slides, screenshots, supplementary files, or spreadsheets. Those inputs can be broken, incomplete, or expressed as salts, charged forms, and tautomers. When analysis tools silently repair structures or spreadsheets silently fill blanks, the student loses the ability to explain how a result was produced.

ProvenanceLab is a rule-based web application that turns a submitted molecular representation into a transparent analysis record. The product validates the input, distinguishes invalidity from ambiguity, standardizes a parent structure while logging every change, calculates descriptors, compares them with frozen reference evidence, and emits a manifest that can be replayed and verified.

0	31	100%	1
silent repairs or filled values	challenge cases in the required test set	missing values carry a reason code	immutable manifest per analysed molecule

1.1 Product principles

- Preserve before transforming. Keep the exact submitted source and structure text.
- Explain, do not merely reject. Pair every issue code with plain-language meaning and a next action.
- Never guess scientific evidence. Distinguish invalid, unavailable, not applicable, outside coverage, and computation failure.
- Require human confirmation. Any structure extracted from an image or document remains an untrusted candidate until a student approves it.
- Pin the environment. Record rules, toolkit versions, snapshot identity, configuration, and hashes.

1.2 Product outcome

OUTCOME

A student can submit a molecule, understand whether it is safe to analyse, see what changed during standardization, compare computed and reference properties without ambiguous blanks, and share a self-describing run record with a classmate or instructor.

2. Problem definition

2.1 Problem statement

How might we help biotechnology students analyse and compare molecules reproducibly, while clearly identifying invalid inputs and missing evidence?

2.2 Why the problem exists

Failure mode	Typical cause	Consequence
Broken structures	Impossible valence, malformed rings, invalid aromaticity, or notation that cannot form a safe molecular graph.	False calculations, parser crashes, or untraceable manual fixes.
Ambiguous structures	Stereocentres, double-bond geometry, protonation, or attachment points are not fully specified.	Different plausible molecules may have different biological effects.
Equivalent forms	Salts, solvates, charged forms, fragments, and tautomers can describe related forms of the same parent.	Identity matching and descriptor values can change after standardization.
Version-sensitive values	Toolkit algorithms and reference releases evolve.	Two analyses may disagree without either person knowing why.
Silent missingness	Spreadsheets and scripts collapse distinct missing states into blank, zero, or N/A.	Students cannot tell whether the value is absent, irrelevant, invalid, or failed.

2.3 Current alternatives and gap

General structure editors are good at drawing molecules; public databases are good at lookup; chemistry toolkits are good at computation; spreadsheets are familiar for comparison. None of these, by default, creates a student-facing chain of custody from source to validation decision to standardized form to evidence-backed result. ProvenanceLab connects those steps and makes uncertainty part of the output contract.

2.4 Constraints

- Analyses must not depend on live PubChem or ChEMBL services.
- Full ChEMBL and PubChem downloads may be too large for the immediate environment; an explicitly scoped subset is acceptable for the first integrated snapshot.
- The chemistry path must remain deterministic and rule based; no model training is required.
- The web application must remain responsive on student laptops and small screens.
- Scientific limitations must be visible in the UI, API, export, and team presentation.

3. Goals, non-goals, and release boundary

ID	Goal	Definition
G1	Precise validation	Convert every accepted input into a valid molecular graph or a deterministic, specific error code.
G2	Student comprehension	Explain the issue, why it matters, the affected location, and the next review step.
G3	Traceable standardization	Create a parent form while preserving before/after representations for every rule.
G4	Honest comparison	Compute descriptors and compare against frozen evidence without synthesizing missing values.
G5	Reproducibility	Make a prior run replayable from inputs, configuration, versions, snapshot, and hashes.
G6	Convenient acquisition	Support direct structure data first, then paper, image, drawing, and batch sources with confirmation.

3.1 Non-goals

- Generating novel molecules, predicting biological activity, or training a chemistry model.
- Automatically correcting invalid structures without an explicit student decision.
- Using live external APIs during a reproducible analysis run.
- Providing clinical, toxicological, regulatory, or therapeutic recommendations.
- Claiming that one standardized parent is the only scientifically meaningful form.
- Replacing the original paper, source file, or laboratory record.

3.2 MVP and target product boundary

Capability	Status	Product requirement
Direct SMILES validation	Working prototype	Ship as the initial end-to-end path.
RDKit depiction and issue location	Working prototype	Harden atom/bond highlighting and invalid-preview language.
Standardization and descriptors	Working prototype	Verify rule ordering and property definitions against the 31 cases.
Manifest and JSON export	Working prototype	Add persistent run storage and replay verification.
Frozen PubChem/ChEMBL evidence	Not configured	Build a versioned, checksum-verified local snapshot adapter.
Paper/photo recognition	Not implemented	Add extraction only with visible confidence and mandatory confirmation.
Drawing editor	Not implemented	Integrate a real chemical editor; do not simulate a canvas.
Multi-molecule comparison UI	Not implemented	Allow add/remove/reorder before batch validation.

4. Users and critical journeys

User	Primary need	Evidence of success
Biotechnology student	Needs to analyse molecules from learning material without becoming a cheminformatics expert.	Clear language, visual location, safe defaults, comparison, downloadable record.
Instructor or reviewer	Needs to understand and verify how students obtained a result.	Stable issue codes, rule log, evidence source, versions, hashes, replay.
Data steward / team developer	Needs to manage snapshots and prevent scientific drift.	Release manifests, checksums, schema validation, regression suite, provenance audit.

4.1 Journey A - direct SMILES

- Student enters an optional label and required SMILES string.
- The application preserves and hashes the exact normalized request payload.
- RDKit returns a depiction plus valid, ambiguous, or invalid status.
- Student confirms the intended structure and opens the analysis record.
- The result explains validation, standardization, properties, evidence, and provenance.

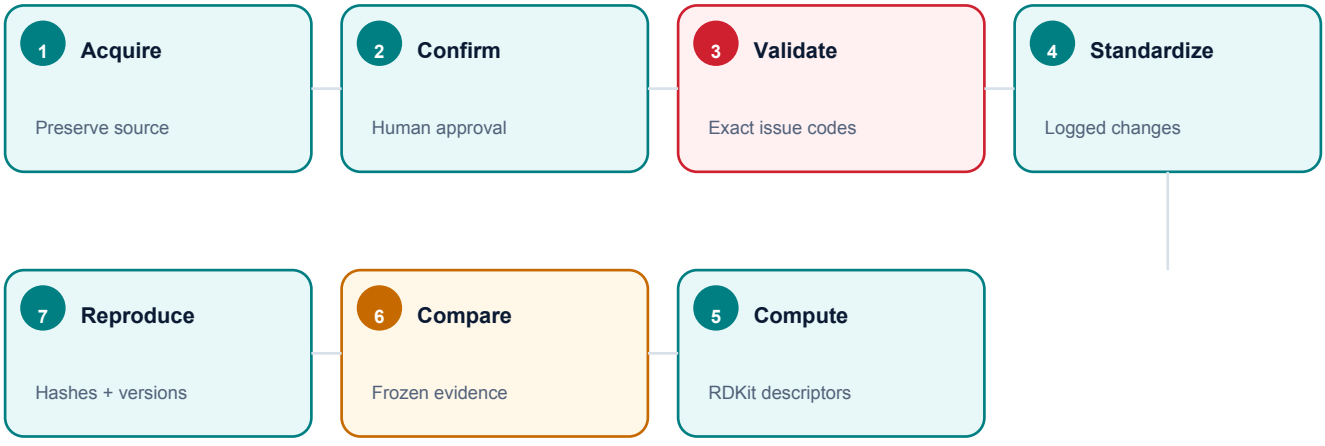
4.2 Journey B - paper or image

- Student uploads a paper, page image, or screenshot; the original file is hashed and preserved.
- Recognition creates one or more untrusted structure candidates with source coordinates and confidence metadata.
- Student accepts, edits, or rejects each candidate; no candidate proceeds silently.
- Confirmed candidates follow the same deterministic chemistry pipeline as direct input.

4.3 Journey C - compare and share

- Student builds a run containing two or more confirmed molecules.
- Invalid molecules remain visible but are excluded from computed comparison cells.
- Every missing cell states its reason; reference and computed values remain separate.
- Student exports a report and machine-readable run record that an instructor can verify.

4.4 Primary pipeline



5. Experience requirements

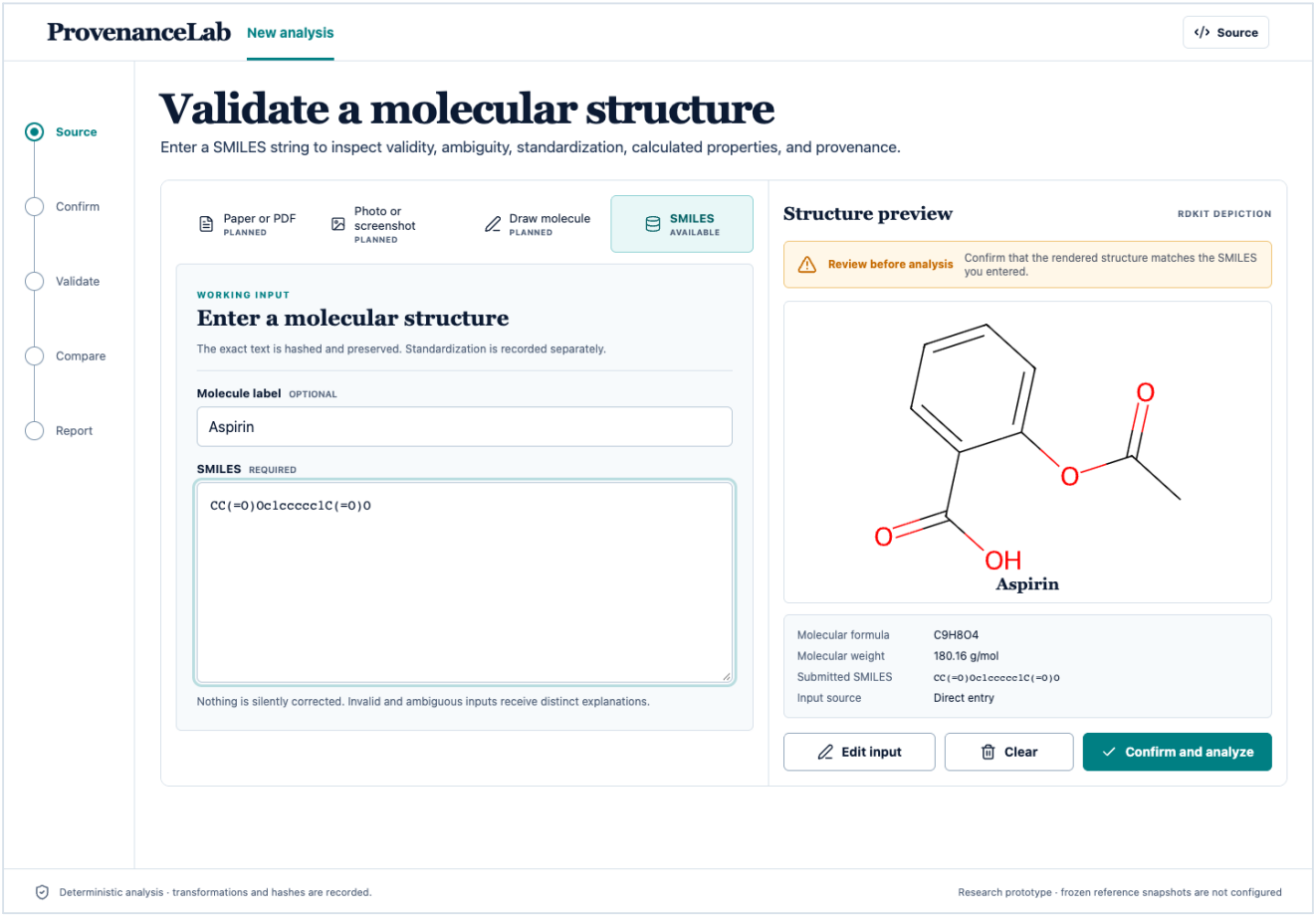


Figure 1. Current direct-SMILES confirmation flow. The displayed content is entered by the user; no sample molecule is preloaded.

5.1 Interaction model

- Start empty. Never pre-populate student data or silently add comparison molecules.
- Label unsupported modes as unavailable or planned; do not present controls that pretend to work.
- Show the molecular depiction before final analysis so the student can confirm identity.
- When preview validation already knows an input is invalid, show the issue immediately and label the action View validation report, not Confirm and analyse.
- Replace generic Unknown with Not computed - invalid input or another explicit reason state.
- Keep raw input, standardized output, computed values, and reference values visually distinct.

5.2 Information hierarchy

Layer	Question answered	Required content
Summary	Can I proceed?	Overall status, count of valid/ambiguous/invalid molecules, primary next action.
Validation	What is wrong or uncertain?	Issue code, title, explanation, affected atoms/bonds, why it matters, next step.
Standardization	What changed?	Ordered rule steps with before/after SMILES and changed flag.
Properties	What was computed?	Value, unit, algorithm context, and non-value reason where applicable.
Evidence	What does the frozen source say?	Snapshot ID, match status, reference value, reason code, source record.
Provenance	Can this be reproduced?	Input/config/output hashes, versions, snapshot, run identifier.

6. Functional requirements

ID	Capability	Requirement	Priority
FR-001	Input preservation	Store the exact submitted structure text and source metadata before any chemistry transformation.	Must
FR-002	Supported inputs	MVP: SMILES. Target: MOL, SDF, CSV, drawing, PDF, PNG, and JPG with type/size validation.	Must
FR-003	Candidate confirmation	Treat extracted/drawn structures as candidates that require accept, edit, or reject.	Must
FR-004	Syntax validation	Return a stable code when notation cannot be parsed into a graph.	Must
FR-005	Chemical validation	Detect impossible valence, sanitization, aromaticity, and related graph failures.	Must
FR-006	Ambiguity detection	Flag unspecified stereochemistry and other supported ambiguity without classifying it as invalid.	Must
FR-007	Issue explanation	Return title, summary, explanation, scientific consequence, next step, technical detail, and location.	Must
FR-008	Depiction	Render valid and partially parsed structures; visually mark affected atoms/bonds when RDKit provides a location.	Must
FR-009	Standardization	Apply cleanup, fragment-parent selection, uncharging, and canonical tautomer rules in a pinned order.	Must
FR-010	Transformation log	Record rule code, label, before/after representation, and whether each step changed the structure.	Must
FR-011	Descriptors	Compute the agreed property set only for chemically valid structures; retain units and algorithm context.	Must
FR-012	Frozen evidence	Resolve reference values only from configured immutable PubChem/ChEMBL snapshots.	Must
FR-013	Missing semantics	Every null value must include a status, reason code, and human-readable reason.	Must
FR-014	Comparison	Display computed and reference values side by side with explicit match/difference/missing status.	Must
FR-015	Batch run	Allow at least 2 and up to 100 structures per API batch; the UI must support add/remove/reorder.	Should
FR-016	Manifest	Return input, configuration, and output hashes with all relevant tool and snapshot versions.	Must
FR-017	Export	Download a machine-readable run record; target release also produces a human-readable PDF report.	Must
FR-018	Replay	Recreate a prior output from its stored input, configuration, toolkit environment, and evidence snapshot.	Must
FR-019	Snapshot administration	Import, validate, index, activate, and retire snapshots without mutating existing run provenance.	Must
FR-020	Health and readiness	Expose service, ruleset, RDKit, and snapshot readiness without reporting unconfigured evidence as healthy.	Should

CRITICAL ACCEPTANCE RULE A parsed drawing is not proof of chemical validity. The preview and final report must present graph depiction and validation status as separate facts.

7. Validation and missing-value contract

7.1 Molecule-level classifications

Classification	Meaning	Property behavior
VALID	No blocking graph error and no supported unresolved ambiguity.	Compute properties; compare when evidence is configured.
AMBIGUOUS	Graph is valid, but one or more chemically relevant details are unspecified.	Compute with a visible warning; preserve the ambiguity in the record.
INVALID	No safe sanitized molecular graph can be produced.	Do not compute properties; evidence lookup is not applicable.

7.2 Property and evidence states

Status	Definition	Display contract
AVAILABLE	A value was computed or found in the configured evidence source.	Value and unit are present.
INVALID_INPUT	The molecular structure is invalid and blocks calculation.	No computed value.
AMBIGUOUS_INPUT	The result is affected by an unresolved input ambiguity.	Value may exist but must carry the warning.
NOT_COMPUTABLE	The property has no computable value for this valid record.	No value; reason required.
NOT_AVAILABLE	The configured system does not provide the value.	No value; reason required.
NOT_APPLICABLE	The calculation or lookup is irrelevant for the current record.	No value; reason required.
NO_REFERENCE_MATCH	A snapshot is active, but no matching compound record was found.	Computed value may remain available.
OUTSIDE_ SNAPSHOT_ COVERAGE	The request is outside the explicitly declared snapshot scope.	Do not imply database absence.
COMPUTATION_ERROR	An expected descriptor calculation failed unexpectedly.	No value; capture toolkit context.
CONFLICTING_EVIDENCE	Computed and reference values differ beyond the configured tolerance.	Show both values and tolerance.

7.3 Initial validation issue codes

Code	Trigger	Required student guidance
SMILES_SYNTAX_ERROR	Parser cannot create a molecular graph.	Check brackets, branches, atom symbols, ring labels, and bond characters.
IMPOSSIBLE_VALENCE	Atom bond order and charge exceed permitted valence.	Review the highlighted atom, charge, and surrounding bonds.
AROMATICITY_ERROR	Aromatic/Kekule representation cannot be resolved.	Review aromatic atom notation and ring bond pattern.
SANITIZATION_ERROR	Graph fails another RDKit sanitization stage.	Show technical detail and avoid guessing a repair.
UNSPECIFIED_STEREOCHEMISTRY	Potential stereocentre is not assigned.	Check wedges, stereochemical marks, or the source record.
MULTIPLE_FRAGMENTS	Input contains disconnected components.	Explain salt/solvent/mixture implications and show parent selection.

8. Standardization and property specification

8.1 Standardization policy

Standardization creates a declared comparison parent; it does not overwrite the submitted molecule. Rules run in a fixed order and each step stores before/after isomeric SMILES plus a changed flag. Any future rule change increments the ruleset version and therefore the configuration hash.

Order	Rule	Purpose	Audit requirement
1	CLEANUP	Normalize supported functional groups and charge patterns.	Record exact RDKit result.
2	FRAGMENT_PARENT	Select the declared parent from disconnected fragments.	Retain all original fragments in raw input.
3	UNCHARGE	Normalize removable formal charges where the rules allow.	Never imply physiological protonation state.
4	CANONICAL_TAUTOMER	Choose the toolkit's canonical tautomer for matching.	Preserve pre-tautomer representation.

8.2 Initial descriptor set

Property	Unit	Method	Initial comparison tolerance
Molecular formula	Text	RDKit molecular formula	Exact equality
Molecular weight	g/mol	RDKit MolWt	0.02
Exact mass	Da	RDKit ExactMolWt	0.001
LogP	Unitless	RDKit Crippen MolLogP	0.15
Topological polar surface area	A^2	RDKit TPSA	0.10
Hydrogen-bond donors	Count	RDKit Lipinski	Exact equality
Hydrogen-bond acceptors	Count	RDKit Lipinski	Exact equality
Rotatable bonds	Count	RDKit Lipinski	Exact equality
Ring count	Count	RDKit Lipinski	Exact equality
Fraction Csp3	Unitless	RDKit fraction Csp3	0.01

DECISION REQUIRED The chemistry lead must approve property definitions and tolerances. A match tolerance is part of the scientific configuration and must be hashed, versioned, and visible in exports.

9. Frozen evidence and data governance

Reproducible evidence comparison requires immutable local data. The product must never use a moving 'latest' directory or a live API during a recorded analysis. Each release is selected deliberately, verified once, transformed deterministically, and then addressed by snapshot ID.

9.1 Source strategy

Source	Recommended acquisition	MVP scope	Required metadata
ChEMBL	Select a numbered release from ftp.ebi.ac.uk/pub/databases/chembl/ChEMBLdb/releases/.	Use a compact SQLite or extracted compound/property subset; avoid the moving latest directory.	Release number, source URL, retrieved time, archive checksum, schema version, licence note.
PubChem	Use NCBI PubChem Compound bulk files from ftp.ncbi.nlm.nih.gov/pubchem/Compound/.	Download only an explicit CID range or curated CID list needed for the test/demo corpus.	CID selection rule, file names, checksums, property fields, retrieval time.

9.2 Snapshot build requirements

- Create a manifest before ingestion: source, release, expected files, size, and published checksums where available.
- Verify all downloads; a failed checksum blocks activation.
- Transform source records with a versioned ingestion program and deterministic ordering.
- Index identifiers needed for parent-form matching: canonical/isomeric SMILES, InChIKey layers where licensed and available, source IDs, and property records.
- Record the transformed database checksum and row counts; make the directory read-only after activation.
- Never overwrite a snapshot. New data creates a new snapshot ID and new configuration hash.

9.3 Evidence matching outcomes

Condition	Status	Required explanation
Snapshot not installed	NOT_AVAILABLE	Reference snapshot is not configured; this is not a database miss.
Molecule outside selected CID/release scope	OUTSIDE_SNAPSHOT_COVERAGE	State the declared coverage boundary.
Within scope but no compound match	NO_REFERENCE_MATCH	No matching snapshot record was found for the standardized identity.
Compound match but property absent	NOT_AVAILABLE	Record exists, but this property is not reported.
Reference present and agrees	AVAILABLE	Show both values, source, and tolerance.
Reference present and differs	CONFLICTING_EVIDENCE	Show both values; do not choose a winner silently.

10. Reproducibility, run records, and API

10.1 Required manifest

Field	Requirement
analysis_id	Human-usable stable identifier derived from input and configuration hashes.
input_sha256	Hash of canonical JSON containing the exact analysis request and source metadata.
configuration_sha256	Hash of pipeline version, ruleset, descriptor definitions/tolerances, and snapshot ID.
output_sha256	Hash of the deterministic result payload, excluding volatile timestamps or UI state.
tool_versions	Pipeline, ruleset, Python, RDKit, ingestion adapter, and relevant OCR/editor versions.
snapshot	Immutable snapshot ID, scope description, source release IDs, and transformed database checksum.
source_hashes	Hashes of original uploaded files when file inputs are supported.
decision_log	Candidate accept/edit/reject events and any user-approved correction.

10.2 Replay contract

- A replay request resolves the exact immutable snapshot and compatible pipeline environment.
- The system reruns from the preserved input, not from the previous standardized output.
- The new output hash must equal the stored output hash; any mismatch is a verification failure with diagnostics.
- If the required environment or snapshot is unavailable, replay returns a blocking missing-dependency status instead of running with replacements.

10.3 API surface

Method	Endpoint	Release	Purpose
GET	/api/health	Current	Service and toolkit versions; target adds snapshot readiness.
POST	/api/analyze	Current	Validate and analyse one SMILES request.
POST	/api/analyze/batch	Current	Analyse 1-100 molecular requests.
POST	/api/sources	Target	Register and hash a document, image, or structure file.
POST	/api/runs	Target	Create a persisted multi-molecule run.
GET	/api/runs/{id}	Target	Retrieve immutable result and manifest.
POST	/api/runs/{id}/replay	Target	Replay and verify against the stored output hash.
GET	/api/snapshots	Target	List configured evidence releases and scope.

11. System architecture and quality requirements

REACT WEB APP	FASTAPI ORCHESTRATION	RDKIT CHEMISTRY	FROZEN DATA
Atomic components, candidate confirmation, result tabs, responsive comparison, export controls.	Typed validation contract, batch coordination, run storage, manifest assembly, error handling.	Parse, sanitize, detect ambiguity, standardize, depict, calculate descriptors.	Versioned PubChem/ChEMBL indexes, checksums, scope manifest, immutable lookup.

11.1 Non-functional requirements

ID	Area	Requirement
NFR-001	Determinism	The same input and configuration in the same pinned environment produce the same output hash.
NFR-002	Performance	Direct SMILES preview p95 <= 1.5 seconds for molecules up to 200 atoms on a reference student laptop.
NFR-003	Batch performance	The 31-case regression corpus completes in <= 10 seconds on the reference development machine.
NFR-004	Offline evidence	Recorded chemistry/evidence analysis makes no live external network request.
NFR-005	Accessibility	Core flow meets WCAG 2.2 AA: keyboard navigation, focus, labels, contrast, and non-colour status cues.
NFR-006	Responsive layout	No shell-level horizontal overflow from 390 px through 1440 px; data tables may scroll within their container.
NFR-007	Security	Validate file type and size, isolate parsing, never execute uploaded content, and exclude secrets from manifests.
NFR-008	Privacy	State storage and retention policy before accepting documents; support deletion of uploaded source data.
NFR-009	Observability	Log request ID, timing, code path, and failure class without logging full sensitive source content by default.
NFR-010	Compatibility	Support current Chrome, Edge, Firefox, and Safari desktop; define one mobile browser baseline.

DEPLOYMENT NOTE

Docker is not required merely because the product is a web app. A container becomes useful when the team needs a repeatable RDKit/runtime/snapshot environment across laptops or deployment hosts. Reproducibility requires an equivalent pinned environment whether delivered by Docker, Conda, lockfiles, or another mechanism.

12. Verification strategy and 31-case corpus

The challenge requires a team-owned set of broken and difficult inputs. Each case must include the raw input, expected molecule classification, issue codes, issue locations where deterministic, standardization expectations, property behavior, evidence behavior, and stable hash expectations.

Category	Cases	Coverage intent
SMILES syntax and ring closure	6	Unclosed rings, branch errors, illegal symbols, bracket mismatch, bond-placement errors.
Valence, sanitization, aromaticity	6	Neutral hypervalence, charge errors, impossible aromatic systems, sanitization failures.
Stereochemistry and geometry	5	Unassigned chiral centres, explicit stereochemistry, double-bond geometry, non-stereogenic marks.
Salts, solvents, and fragments	4	Counterions, solvates, mixtures, and parent selection.
Charges and protonation	3	Removable charges, persistent charges, zwitterions.
Tautomers and equivalent forms	3	Canonical tautomer stability and before/after audit.
Evidence and missing semantics	4	No snapshot, outside scope, no match, missing property, and conflict across the four cases.

12.1 Required automated checks

- Unit tests for each validation issue-code path and standardization rule.
- Golden contract tests for the 31 cases, reviewed by the chemistry owner.
- Hash determinism test across repeated runs in the same environment.
- Snapshot ingestion tests for checksum failure, schema mismatch, duplicate records, and row-count drift.
- API schema tests for every status and null reason.
- Browser tests for valid, ambiguous, invalid, backend-unavailable, no-snapshot, and empty-session states.
- Responsive and accessibility checks at desktop and 390 px mobile widths.

12.2 Acceptance gates

Gate	Pass condition
Scientific correctness	31/31 cases match the approved molecule classification and issue-code contract.
No silent missingness	100% of null property/evidence values include status, reason code, and explanation.
Reproducibility	Three repeated runs per case produce identical input, configuration, and output hashes.
Evidence integrity	Snapshot mismatch or absent snapshot blocks reference values and is visible in UI/export.
UX clarity	A student can distinguish invalid, ambiguous, unavailable, and not applicable without opening developer tools.
Regression safety	Backend tests, frontend lint/build, and browser smoke tests pass before release.

13. Current implementation assessment

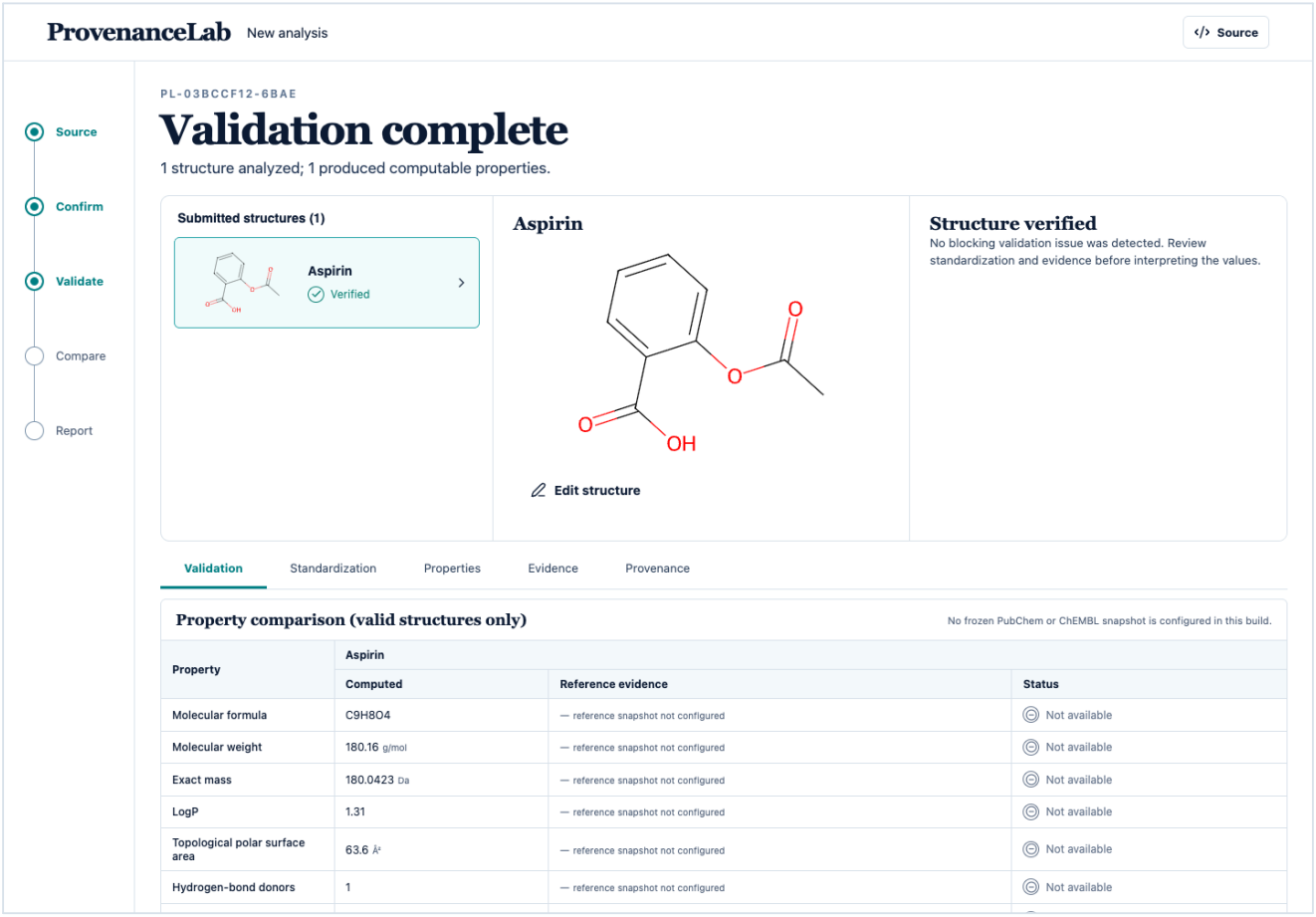


Figure 2. Current valid-result screen. Computed values are available; reference cells explicitly state that no frozen snapshot is configured.

13.1 What is functional

- React/TypeScript UI with an atomic component hierarchy and lazy-loaded result route.
- Direct SMILES input, empty initial state, RDKit SVG preview, and student confirmation.
- FastAPI endpoints for one molecule, batches up to 100, and service health.
- Specific syntax, impossible-valence, aromaticity, sanitization, stereochemistry, and fragment issues.
- Deterministic standardization log and ten computed descriptors.
- Explicit unconfigured-evidence status, manifest hashes, toolkit versions, and JSON export.
- Verified desktop/mobile rendering with no relevant console errors in the tested flows.

13.2 Known gaps

- No frozen PubChem or ChEMBL snapshot is installed; evidence comparison is correctly unavailable.
- Paper/photo recognition, structure-file upload, and drawing are not implemented.
- The UI accepts one molecule per run even though the API supports a batch.
- Run state is session-only; previous runs cannot yet be retrieved or replayed after storage loss.
- Invalidity is detailed after confirmation; the preview should surface the known error and explicit non-computation reason earlier.
- The required 31-case corpus is not yet fully authored.

14. Delivery plan, ownership, and success

14.1 Milestone plan

Milestone	State	Exit outcome
M0 - Honest prototype	Complete	Direct SMILES flow, core validation, standardization, descriptors, manifest, clear unsupported modes.
M1 - Chemistry contract	Next	Approve rules/tolerances, author 31 cases, improve preview invalid-state semantics, reach acceptance gates.
M2 - Frozen evidence	Next	Choose releases/subset, build checksum manifest and ingestion adapters, expose match provenance.
M3 - Comparison and persistence	Planned	Multi-molecule UI, immutable run storage, report export, exact replay verification.
M4 - Convenient acquisition	Planned	MOL/SDF/CSV, drawing editor, paper/image extraction, candidate source coordinates.
M5 - Release hardening	Planned	Accessibility, cross-browser, security, performance, documentation, deployment package.

14.2 Suggested owners

Workstream	Primary owner	Immediate deliverable
Chemistry rules	Chemistry/backend	Approve issue taxonomy, rule order, descriptor definitions, and 31 expected outputs.
Snapshot pipeline	Data/backend	Choose ChEMBL release and PubChem CID scope; create manifest and checksum verifier.
Student experience	Frontend/UX	Surface invalid reasons in preview; add multi-molecule input and explicit evidence states.
Reproducibility	Backend/platform	Persist immutable run payloads and implement replay/hash verification.
Quality	QA + chemistry reviewer	Automate golden corpus, API schema, browser, responsive, and accessibility checks.
Demo and documentation	Product/presentation	Maintain the status matrix, limitations, demo script, and teammate handoff materials.

14.3 Success measures

- 100% of approved invalid cases produce the expected stable code and block descriptor calculation.
- 100% of missing values expose the correct reason category and explanation.
- 100% of replayable runs verify the stored output hash when dependencies are present.
- At least 90% of usability-test students correctly explain why an invalid, ambiguous, and unavailable result differs.
- At least 95% of usability-test students complete the direct-SMILES flow without assistance.
- p95 preview latency meets the 1.5-second target on the reference student laptop.

15. Risks, decisions, and definition of done

15.1 Risk register

Risk	Severity	Failure	Mitigation
Scientific overclaim	High	UI may imply that a parsed depiction is valid or that a normalized parent is the original molecule.	Separate depiction from validation; preserve raw input; chemistry review of copy.
Snapshot scale	High	Full archives exceed available time/storage.	Start with declared CID/property scope; expose outside-coverage status.
Cross-version drift	High	RDKit or ingestion updates change values/hashes.	Pin environment and versions; golden tests; explicit migration rather than overwrite.
OCR false confidence	High	Paper/image recognition extracts the wrong structure.	Untrusted candidates, confidence/source location, mandatory human confirmation.
Identity mismatch	Medium	Parent/tautomer rules collapse scientifically distinct forms.	Retain all representations; version rules; show transformations and source form.
Privacy	Medium	Uploaded papers may contain unpublished or licensed content.	Define retention, local processing, access controls, and deletion before enabling upload.

15.2 Decisions required from the team

- Which numbered ChEMBL release and PubChem CID/property subset define the first evidence snapshot?
- Which input formats are required for the judged demo versus the product roadmap?
- Which standardization rules and comparison tolerances receive chemistry-owner approval?
- Will run artifacts be local-only, stored in a shared backend, or both? What is the retention policy?
- Which drawing component and document-recognition approach meet licensing, offline, and confirmation needs?
- Who signs off the 31-case expected outputs and any future ruleset migration?

15.3 Definition of done for the first shareable release

- All Must requirements in Section 6 are implemented or explicitly scoped out of the release with product-owner approval.
- A numbered frozen evidence snapshot is active, checksum-verified, and visible in every run manifest.
- The 31-case golden corpus passes with no silent nulls or placeholder UI values.
- A previous run can be retrieved, replayed in its required environment, and verified by output hash.
- Desktop and mobile core flows pass accessibility, browser, performance, and error-state checks.
- The README, API contract, PRD, demo script, and limitations all describe the same capability boundary.

TEAM ALIGNMENT CHECKPOINT

Approve the first snapshot scope, 31-case ownership, standardization policy, and persistence model before expanding the input surface. Those four decisions determine the scientific contract and prevent expensive UI work from getting ahead of the evidence pipeline.

16. Reference links and glossary

16.1 Official resources

Resource	Location	Use in ProvenanceLab
Project repository	https://github.com/Office50505/32-Bit	Source code, README, verified screenshots, and implementation history.
RDKit	https://www.rdkit.org/	Parsing, sanitization, standardization, depiction, and molecular descriptors.
ChEMBL numbered releases	https://ftp.ebi.ac.uk/pub/databases/chembl/ChEMBLdb/releases/	Select and pin one numbered release for frozen evidence.
ChEMBL download documentation	https://chembl.gitbook.io/chembl-interface-documentation/downloads	Understand release formats and supported download options.
PubChem Compound bulk files	https://ftp.ncbi.nlm.nih.gov/pubchem/Compound/	Acquire an explicitly scoped CID/property subset for the frozen snapshot.

16.2 Glossary

Term	Working definition
SMILES	A line notation that represents atoms, bonds, branches, rings, charges, and optional stereochemistry.
Sanitization	Toolkit checks that establish chemically consistent valence, aromaticity, charge, and related graph properties.
Ambiguity	A valid graph whose scientifically relevant identity is not fully specified, such as an unassigned stereocentre.
Parent form	The standardized component selected for matching and comparison while the submitted multicomponent form remains preserved.
Tautomer	One of several interconvertible structures that differ in proton placement and bond arrangement.
Frozen snapshot	An immutable, checksum-verified local dataset with a declared source release and coverage boundary.
Provenance manifest	The machine-readable record of hashes, configuration, tool versions, snapshot identity, and source metadata.
Replay	Re-execution from the preserved input and exact configuration followed by output-hash verification.

DOCUMENT CONTROL Update this PRD whenever the ruleset, status taxonomy, snapshot policy, descriptor set, or release boundary changes. Product copy, API contracts, tests, and demonstrations must remain aligned with the approved revision.